Extra Dessert #21: Antenna Figure and Pattern Drill

Every antenna figure and pattern type the exam tests, with the values to memorize and the cues to recognize each pattern by name.

Supports EC-C, B2, B9, B16, B18 · Self-study reference · Practice exams at eham.net/exams

Official Exam Figures: Memorize These Numbers

The pool has two antenna-pattern figures, both tested in E9B. Numbers come straight off the polar plots.

Figure	What it shows	Numbers to memorize
Figure E9-1	Azimuth pattern (top-down view).	**3 dB beamwidth = 50°**. **Front-to-back = 18 dB**. **Front-to-side = 14 dB**.
Figure E9-2	Elevation pattern (side view).	Pattern type = **elevation**. **Peak elevation angle = 7.5°**. **Front-to-back = 28 dB**.

How to read a polar pattern: the outer ring is the antenna's maximum response. Each concentric ring is a fixed dB step (typically 5 or 6 dB). Count rings from the peak to the feature you want. F/B is rings between peak (0°) and direct-rear (180°); F/S is peak vs the 90° point; beamwidth is the angular width where the lobe has dropped 3 dB.

Two Verticals: Three Pattern Recipes (Bite 2)

Take two quarter-wave vertical antennas; vary spacing and feed phase to get three named patterns.

Configuration	Pattern	Where the lobes go	Q
1/2 λ apart, 180° out of phase	**Figure-8 end-fire**	Two lobes along the line through the elements.	E9C01
1/4 λ apart, 90° out of phase	**Cardioid**	One forward lobe, deep null behind.	E9C02
1/2 λ apart, in phase	**Figure-8 broadside**	Two lobes perpendicular to the line through the elements.	E9C03

Mnemonic: half-wave + 180° → end-fire (waves cancel sideways, reinforce along the line). Half-wave + in-phase → broadside (the opposite). Quarter-wave + 90° → heart-shaped cardioid.

Pattern Vocabulary Drill

Cover the right column and try to define each term cold.

Term	Definition
Main lobe	The largest bulge in the pattern – the direction where most of your power goes.
Side lobe	A smaller bulge off to the side of the main lobe.
Back lobe	The response at 180° from the main lobe.
Null	An angle where the antenna response drops to (or near) zero. The curve pinches in.
3 dB beamwidth	The angular width of the main lobe measured at the points where the response is 3 dB below the peak.
Front-to-back ratio (F/B)	Main lobe response minus the response 180° behind, in dB.
Front-to-side ratio (F/S)	Main lobe response minus the response at 90° off the main direction, in dB.
Takeoff angle	Vertical angle at which the main lobe leaves the antenna. Low = better DX.
Broadside	Perpendicular (90°) to the line through the antenna elements.
End-fire	Along the line through the antenna elements.
Cardioid	Heart-shaped pattern: one strong forward lobe, single deep null behind.

Ground And Sky Effects

Situation	Effect on takeoff angle / radiation	Q

Vertical antenna over seawater vs soil	Low-angle radiation **increases** (seawater is a near-perfect mirror).	E9C11
Raise a horizontal antenna higher above ground	Lowest-lobe takeoff angle **decreases**.	E9C13
Horizontal antenna over a downhill slope	Main lobe takeoff angle **decreases in the downhill direction**.	E9C14

Antenna Type Recognition Drill

Antenna	Quick recognition fact	Q
Folded dipole	Half-wave dipole + parallel wire connecting the ends. **300 Ω feed.**	E9C07, E9C08
Off-center-fed dipole	Fed away from center. Same Z on several harmonically related bands.	E9C05
G5RV	Center-fed wire (102 ft) + specific open-wire matching section + balun + coax.	E9C09
Zepp	End-fed half-wavelength dipole.	E9C10
Extended double Zepp	Center-fed **1.25 λ** dipole. Narrower main lobe, some forward gain.	E9C12
Yagi driven element	About **1/2 λ** long.	E9D05
Yagi parasitic elements	Longer or shorter than resonance → **phase shift control** (reflector vs director).	E9D12
2-element Yagi with reflector (vs director)	Reflector gives **higher gain** at normal spacing.	E9D11
Circular polarization from two Yagis	Same axis, perpendicular elements, same boom point, fed **90° out of phase**.	E9D02
Parabolic dish	Double the frequency → **+6 dB gain** (ideal dish).	E9D01
Loading coil on short whip	Best location: **center of the radiator**.	E9D03
Beverage	Receiving antenna, **at least 1 λ long**. Termination resistor absorbs reverse-direction signals.	E9H01, E9H07
Small DF loop	**Bidirectional null** – two nulls 180° apart.	E9H05
Pennant (single-turn terminated loop)	**Cardioid** receiving pattern.	E9H09

Matching System Recognition Drill (Yagi And Tower)

Match	Cue	Q
Gamma	Coax shield to center of element; center conductor offset. Series capacitor cancels inductive reactance. Element bonded to boom.	E9E02, E9E04
Beta (hairpin)	Shunt inductor at feed. Element **insulated from boom**. Feed Z must be **capacitive**.	E9E01, E9E05
Stub	Short length of line in parallel near the feed point. Reactance from the stub cancels antenna reactance.	E9E03
Q-section	Quarter-wave line, Z_Q = sqrt(Z1 × Z2). 100 Ω → 50 Ω: ideal Z_Q ≈ 70.7 Ω; use **75 Ω** coax (closest standard).	E9E06
Shunt-feed a grounded tower	Use the **gamma** match.	E9E09

Drill discipline: If you can read both official figures and name all the pattern recipes from memory, the E9C and E9B groups are essentially free points. The “recipe” pages (two verticals, named wire antennas, matching systems) are the highest-leverage memorization in the antenna subelement.